

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
NETWORK SECURITY (CYS360)
(6TH SEMESTER, 3RD Year) LAB EXPERIMENT # 2

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Score:	_____	Date:	_____	Instructor's Signature:	_____

Lab # 2: To Configure Default Route and Packet Filtering Rules on Firewall

Objective: The objective of this lab to Configure Default Route and Packet Filtering Rules on Firewall.

Duration: 3 Hours

Expected Outcomes: By the end of this lab session, students will be able to:

1. Configure basic system settings on the firewall, including device hostname and system clock.
2. Assign IP addresses and security zones to firewall interfaces and ensure proper zone deployment.
3. Establish Internet connectivity by configuring a default route and enabling basic packet-filtering policies.

Rubrics

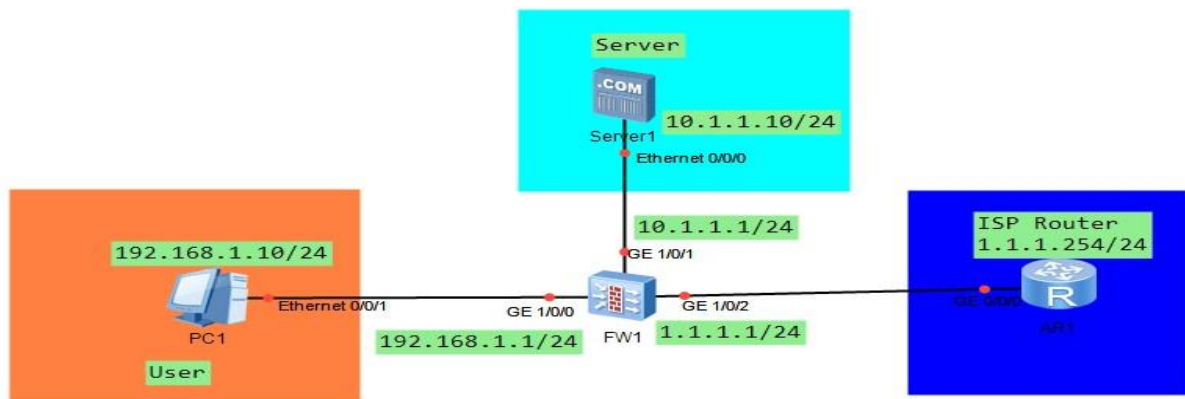
LAB PERFORMANCE INDICATOR	SUBJECT KNOWLEDGE	DATA ANALYSIS AND INTERPRETATION	ABILITY TO CONDUCT EXPERIMENT
SCORE			

Note: Student performance will be assessed based on this rubric during lab sessions.

Scenario

As shown in Figure 1, an enterprise deploys a FW as the enterprise network egress gateway. After logging in to the FW, the administrator needs to complete basic network configuration on the FW, including setting the device name, clock, and interface IP addresses, adding interfaces to security zones, and configuring default routes and default packet filtering rules.

Topology



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Figure 1 Networking diagram for basic network configuration

Configuration Roadmap

1. Set a device name for the FW.
2. Set the clock on the FW.
3. Set IP addresses for the service interfaces on the FW. The IP addresses must be planned in advance.
4. Add the service interfaces to security zones. Generally, the interfaces connecting to the Internet must be added to security zones with a low security level (such as the Untrust zone), and the interfaces connecting to the enterprise network must be added to security zones with high security level (such as the Trust zone). Servers can be deployed in the DMZ.
5. Configure a default route and set the next hop to the access point provided by the ISP.
6. Configure the default packet-filtering rule to permit the FW to access the Internet. By default, the FW is prohibited from accessing the Internet. **Configuration Procedure on the CLI**

1. Set a device name for the FW.

```
<FW> system-view
[FW] sysname FW_A
[FW_A] quit
```

2. Set the time and time zone.

```
<FW> clock datetime 18:10:45 2014/01/01
<FW> clock timezone BJ add 08:00:00
```

Beijing is in time zone UTC+8. Therefore, you need to add 8 hours to the default UTC time zone during configuration.

3. Set interface IP addresses.

The IP addresses (1.1.1.1/24 in this example) of the interfaces connecting to the ISP network needs to be obtained from the ISP.

```
<FW_A> system-view
[FW_A] interface GigabitEthernet 1/0/0
[FW_A-GigabitEthernet1/0/0] ip address 192.168.1.1 24
[FW_A-GigabitEthernet1/0/0] quit
[FW_A] interface GigabitEthernet 1/0/1
[FW_A-GigabitEthernet1/0/1] ip address 10.1.1.1 24
[FW_A-GigabitEthernet1/0/1] quit
[FW_A] interface GigabitEthernet 1/0/2
[FW_A-GigabitEthernet1/0/2] ip address 1.1.1.1 24
[FW_A-GigabitEthernet1/0/2] quit
```

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4. Add the service interfaces to security zones.

```
[FW_A] firewall zone trust
[FW_A-zone-trust] add interface GigabitEthernet 1/0/0
[FW_A-zone-trust] quit
[FW_A] firewall zone dmz
[FW_A-zone-dmz] add interface GigabitEthernet 1/0/1
[FW_A-zone-dmz] quit
[FW_A] firewall zone untrust
[FW_A-zone-untrust] add interface GigabitEthernet 1/0/2
[FW_A-zone-untrust] quit
```

5. Configure a default route.

```
[FW_A] ip route-static 0.0.0.0 0.0.0.0 1.1.1.254
```

6. Configure the default packet-filtering rule to permit the FW to access the Internet.

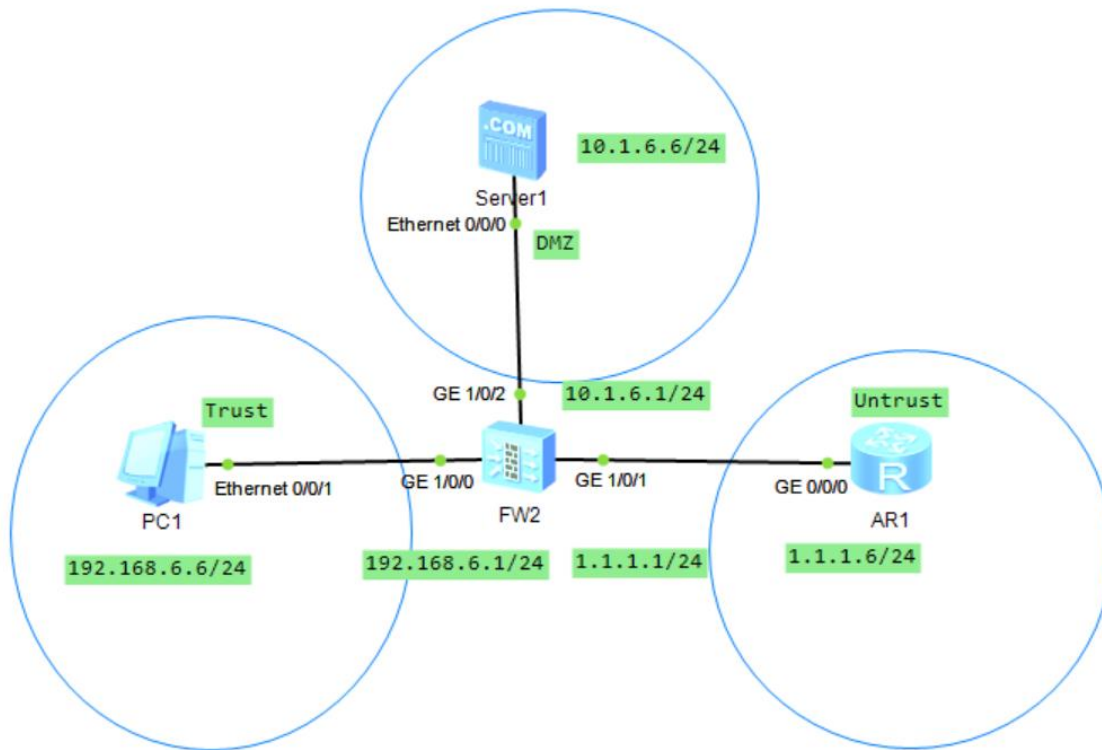
```
[FW_A] security-policy
[FW_A-policy-security] default action permit
```

Note: Attach all the screenshots of the configuration. The Firewall name should be your roll number like FW1-23BSCYS001. Upload the .topo and .cfg files on MS Teams Assignment.

Task:

- 1. Check the connectivity between PC1 and Server and attach the screenshot. You will get the ping reply.**
- 2. Check the connectivity between PC1 and ISP Router and attach the screenshot.**

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```
[FW-23BSCYS006]dis zone interface
2025-12-29 05:38:26.400 +05:00
local
  interface of the zone is (0):
#
trust
  interface of the zone is (2):
    GigabitEthernet0/0/0
    GigabitEthernet1/0/0
#
untrust
  interface of the zone is (1):
    GigabitEthernet1/0/1
#
dmz
  interface of the zone is (1):
    GigabitEthernet1/0/2
#
[FW-23BSCYS006]|
```

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```
[FW-23BSCYS006]dis security-policy rule
[FW-23BSCYS006]dis security-policy rule all
2025-12-29 05:39:10.110 +05:00
Total:3

```

RULE ID	RULE NAME	STATE	ACTION	HITS
1	trust-to-untrust	enable	permit	8
2	untrust-to-trust	enable	deny	0
0	default	enable	permit	5

```
[FW-23BSCYS006]|
```

PC1

Basic Config Command MCPacket UdpPacket Console

Welcome to use PC Simulator!

PC>ping 10.1.6.6

Ping 10.1.6.6: 32 data bytes, Press Ctrl_C to break

Request timeout!

From 10.1.6.6: bytes=32 seq=2 ttl=254 time<1 ms

From 10.1.6.6: bytes=32 seq=3 ttl=254 time=16 ms

From 10.1.6.6: bytes=32 seq=4 ttl=254 time<1 ms

From 10.1.6.6: bytes=32 seq=5 ttl=254 time<1 ms

--- 10.1.6.6 ping statistics ---

5 packet(s) transmitted

4 packet(s) received

20.00% packet loss

round-trip min/avg/max = 0/4/16 ms

PC>ping 1.1.1.6

Ping 1.1.1.6: 32 data bytes, Press Ctrl_C to break

Request timeout!

Request timeout!

Request timeout!

Request timeout!

Request timeout!

--- 1.1.1.6 ping statistics ---

5 packet(s) transmitted

0 packet(s) received

100.00% packet loss

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Questions:

1. Purpose of configuring a default route on a firewall

To forward traffic to a next-hop gateway when no specific route exists for the destination network.

2. Difference between a default route and specific route entries

A default route (0.0.0.0/0) matches all destinations, while specific routes match particular networks and have higher priority.

3. Typical steps to set a default route on a firewall

Define the default network (0.0.0.0/0) and configure the next-hop IP or outbound interface.

4. How to verify correct default route configuration

Use routing table display commands and test connectivity (e.g., ping or traceroute).

5. Difference between stateful and stateless packet filtering

Stateful filtering tracks connection states, while stateless filtering checks packets independently.